Analysis of Continuity of Real-valued functions

Let (X, d) be a metric space and $f : X \to \mathbb{R}$ be a function. Let E_f (or simply E) be the set of points of discontinuity of f, i.e., $E_f = \{x \in X : f \text{ is not continuous at } x\}$. Then $X \setminus E$ is the set of points of continuity of f.

The aim of this note is to show that $X \setminus E$ is a G_δ-set.

Definition : A countable intersection of open sets in a metric space is called a G_δ-set. On the other hand, a countable union of closed sets is called an F_σ-set.

Examples : Consider the real line $\mathbb{R}$. Every countable subset of $\mathbb{R}$ is an F_σ-set as singleton subsets are closed. In particular, the set $\mathbb{Q}$ of rationals is an F_σ-set. Since $\mathbb{Q}$ is countable, we see that the set $\mathbb{R} \setminus \mathbb{Q}$ of irrationals is a G_δ-set. In fact,

$$\mathbb{R} \setminus \mathbb{Q} = \bigcap_{p \in \mathbb{Q}} (\mathbb{R} \setminus \{p\}).$$

Question : Is $\mathbb{Q}$ a G_δ-set in the real line? (Answer : No!)

Let $x \in X$ and $\delta > 0$ be given. Let $B(x; \delta)$ be a ball in (X, d). Consider

$$\Omega_f(x, B(x; \delta)) = \sup\{|f(u) - f(v)| : u, v \in B(x; \delta)\}$$

and

$$\omega_f(x) = \inf\{\Omega_f(x, B(x; \delta)) : \delta > 0\}.$$

We say that $\omega_f(x)$ is the *oscillation* of f at the point $x \in X$.

Examples : Let $f : \mathbb{R} \to \mathbb{R}$ be a function.

1. If f is the Dirichlet function, i.e., $f = \chi_{\mathbb{Q}}$ is the characteristic function of $\mathbb{Q}$, then $\omega_f(x) = 1$ for all $x \in \mathbb{R}$. Also, $E_f = \mathbb{R}$.

2. Suppose $f(x) = x$ if $x \in \mathbb{Q}$ and $f(x) = 0$ for $x \in \mathbb{R} \setminus \mathbb{Q}$. Then $\omega_f(0) = 0$ but $\omega_f(x) = x$ for $x \neq 0$. Further, $E_f = \mathbb{R} \setminus \{0\}$.

3. Suppose $f(x) = \lfloor x \rfloor$. Then $\omega_f(x) = 0$ for $x \in \mathbb{R} \setminus \mathbb{Z}$ and $\omega_f(x) = 1$ if $x \in \mathbb{Z}$. In this case, $E_f = \mathbb{Z}$.

4. Suppose $f(x) = \sin(\frac{1}{x})$ for $x \neq 0$ and $f(0) = 0$. Then $\omega_f(0) = 2$ and $\omega_f(x) = 0$ for $x \neq 0$. Also, we have $E_f = \{0\}$.

5. Suppose f is the *Thomae function* (also called *Modified Dirichlet function*). In fact, if $x \in \mathbb{Q}$ and $x = \frac{p}{q}$ in standard form, then $f(x) = \frac{1}{q}$. For irrational x, $f(x) = 0$. We have $\omega_f(\frac{p}{q}) = \frac{1}{q}$, while for irrational x, $\omega_f(x) = 0$. Here $E_f = \mathbb{Q}$.

Proposition : The function $\omega_f : X \to \mathbb{R}$ given by $x \mapsto \omega_f(x)$ is an upper-semicontinuous function. In other words, $(\omega_f)^{-1}((-\infty, b))$ is an open set in (X, d) for every $b \in \mathbb{R}$.

Proof : Since $\omega_f(x) \geq 0$ for all $x \in X$, we need to take $b > 0$. Also,

$$(\omega_f)^{-1}((-\infty, b)) = (\omega_f)^{-1}([0, b)).$$

If $y \in (\omega_f)^{-1}((-\infty, b))$, then $0 \leq \omega_f(x) < b$. Thus, $\exists\ \delta > 0$ such that

$$\Omega_f(y, B(y : \delta)) < b.$$

Now for every $x \in B(y; \frac{\delta}{2})$, clearly $B(x; \frac{\delta}{2}) \subset B(y; \delta)$. Hence,

$$\Omega_f(x, B(x; \frac{\delta}{2})) \leq \Omega_f(y, B(y; \delta)) < b$$

and we have $\omega_f(x) < b$ for every $x \in B(y; \frac{\delta}{2})$. $\square$

Proposition : The function $f : X \to \mathbb{R}$ is continuous at $x \in X$ if and only if $\omega_f(x) = 0$.

Proof: Let f be continuous at $x \in X$. Then for every $\epsilon > 0$, $\exists\ \delta > 0$ such that

$$f(B(x; \delta)) \subseteq (f(x) - \frac{\epsilon}{2}, f(x) + \frac{\epsilon}{2}).$$

Thus for any $u, v \in B(x; \delta)$, $|f(u) - f(v)| < \epsilon$. This shows that $\Omega_f(x, B(x; \delta)) \leq \epsilon$ and $0 \leq \omega_f(x) \leq \epsilon$. Letting ϵ tends to 0, we get $\omega_f(x) = 0$.

Conversely, suppose $\omega_f(x) = 0$. Then for a given $\epsilon > 0$, there exists $\delta > 0$ such that $\Omega_f(x, B(x; \delta)) < \epsilon$. Now for $u \in B(x; \delta)$, we have $|f(u) - f(x)| < \epsilon$. This shows that f is continuous at the point x. $\square$

Proposition : The set of points of continuity of f is a G_δ-set, i.e., $X \setminus E_f$ is a G_δ-set.

Proof : From the last Proposition, $x \in X \setminus E_f \iff \omega_f(x) = 0$. Since

$$\{0\} = \bigcap_{n=1}^{\infty} [0,\ \frac{1}{n}),$$

and ω_f is upper-semicontinuous, we have

$$X \setminus E_f = (\omega_f)^{-1}(\{0\}) = \bigcap_{n=1}^{\infty} (\omega_f)^{-1} \left([0,\ \frac{1}{n}) \right).$$

Hence, $X \setminus E_f$ is a G_δ-set. $\square$

Exercise : Show that every open interval in the real line is an F_σ-set, while every closed interval is a G_δ-set. Further, show that a countable union of F_σ-sets is F_σ and a countable intersection of G_δ-sets is G_δ.